Rule-based Monetary Policy in China

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Motivation and Research Question

- **Motivation**
  - China: The most significant economy but its monetary policy remains under-studied.
  - Currently facing a severe post-COVID demand slowdown.
    - Housing market slump; Local government debt crisis.
    - CPI stagnating around 0%; PPI negative since late 2022.
  - The People's Bank of China (PBoC)'s policy response appears passive or constrained.
    - No explicit monetary policy rule to anchor expectations.
    - Gradual and limited cuts to interest rates and reserve requirements ratio.
- **Research Question:**
  - **How would the Chinese Economy have differed if the PBoC had followed an explicit, rule-based (e.g., NGDP and Inflation Targeting) monetary policy?**

Empirical Strategy

- **Methodology:** Use the **semi-structural method** (McKay & Wolf 2023 ; Caravello & McKay & Wolf 2025) to simulate counterfactual monetary policy rules in China.
- This approach requires two "sufficient statistics" instead of a full structural model, making it robust to the Lucas Critique.
- **Statistic 1: Reduced-Form Projections**
  - **What it is:** The economy's baseline forecast, capturing all "fundamental" shocks (fiscal, demand, etc.) without needing to identify them individually.
  - **Estimation:** We will estimate this using a **Reduced-Form VAR** on key Chinese macro variables.
  - **Time Period**: Quarterly data from 2015 to 2025; for robustness, we will also use 2020-2025.

Empirical Strategy

- **Statistic 2: Policy Causal Effects**
  - **What it is:** The causal effect (IRF) of PBoC's policy tools.
  - **Estimation Method:** We will use a High-Frequency Identification (HFI) approach, following Das and Song (2022, IMF WP).
  - **Identification:** They defined a "shock" as the change in 1-year Interest Rate Swaps (IRS) on the exact day of a PBoC policy announcement.
  - **Policy Tools:**
    - **Price-based tools:** Medium-term Lending Facility (MLF) and Loan Prime Rate (LPR) announcements.
    - **Quantity-based tools:** Required Reserve Ratio (RRR) changes.
- **Simulation of Counterfactuals:**
  - The simulation solves for the optimal sequence of policy interventions (Statistic 2) required to steer the economy's baseline forecast (Statistic 1) to the new counterfactual target (Inflation, NGDP, etc.).

Appendix

The Semi-Structural Method for Policy Counterfactuals

- **The Goal:** Evaluate policy counterfactuals (e.g., "what if the Fed had a different rule?") without committing to a fully specified structural model.
- **Core Insight:** A change in the policy *rule* is equivalent to an appropriate, expected path of policy *shocks* applied to the *baseline* economy forecast.
- **Surviving the Lucas Critique:**
  - The method models a systematic, fully expected rule change, not a sequence of surprises.
  - It uses a combination of date-0 shocks (contemporaneous + "news" shocks) to enforce the new rule in agents' expectations from the start.
- **Key "Structural" Assumption (The Split):**
  - The economy can be split into a **"non-policy block"** (private sector behavior) and a **"policy block"** (the rule).
  - The non-policy block is assumed to be stable and depends only on the *path of the policy instrument* (e.g., the yield curve), not on the rule that generates it.

Counterfactual Rules

1. **Strict Inflation Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 \right]$$

2. **Strict NGDP Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\Delta x_t - \Delta x_t^*)^2 \right]$$

3. **Flexible Inflation Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 + \lambda_y (y_t - y_t^*)^2 \right]$$

4. **Triple Mandate (PBoC-style):**
   - Includes the PBoC's explicit goal of currency stability.

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 + \lambda_y (y_t - y_t^*)^2 + \lambda_e (e_t - e_t^*)^2 \right]$$

Sample Results from the Great Recession

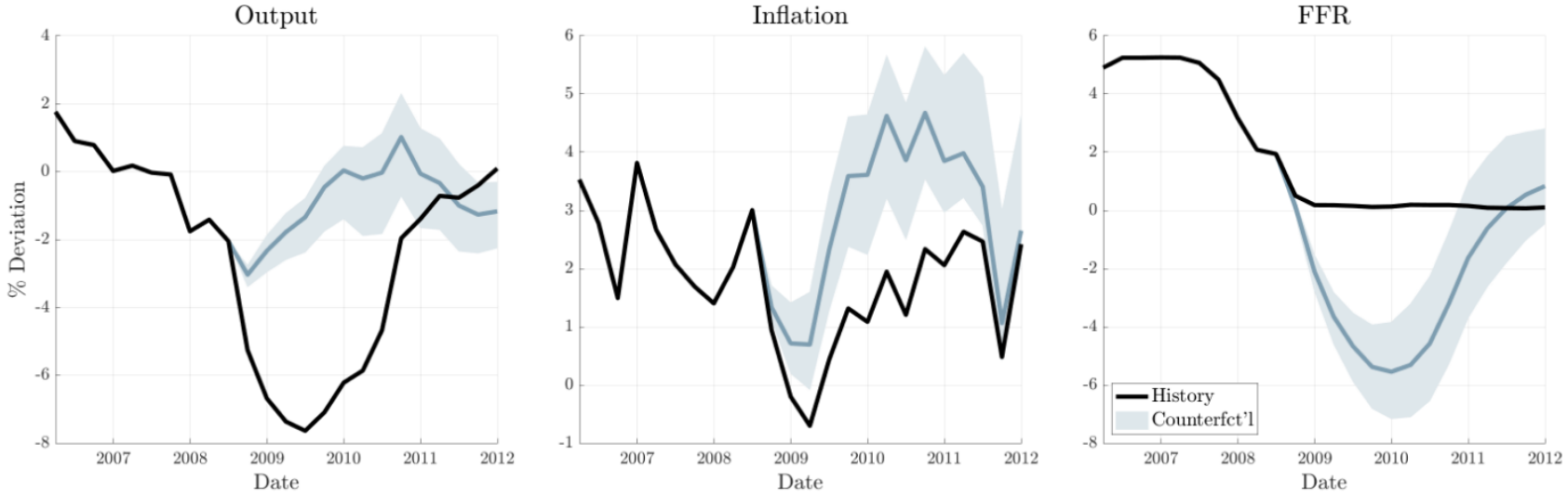

Figure 9: Counterfactual evolution of output, inflation, and the federal funds rate in the Great Recession, under the policy rule that minimizes (9) without any effective lower bound on rates. Black: data. Blue: posterior median (solid) and 16th and 84th percentile bands (shaded), drawing from posterior across all models and parameters.

Figure: Counterfactual Simulations of a flexible inflation targeting rule during the Great Recession (Caravello, McKay, Wolf 2025)

The Structural Part

- **Robustness Check / Future Work (The "Hybrid" Method):**
  - **Problem:** If our baseline fails (or as a check), we can address the critique that the empirical "Recipe" is incomplete (i.e., missing "long-end" effects).
  - **Check:** We can use a minimal structural model (e.g., a simple NK model) and "discipline" it by forcing its short-term IRF to match our HFI-estimated IRF but do not model shocks and monetary policy rules.
  - We then use this "disciplined" model to extrapolate the long-term part of our "Recipe" and re-run the simulation.

How to Measure Chinese Output

- **The Challenge:** Official quarterly GDP data is famously smoothed, low-frequency and over-revised.
- **Primary Approach:** We may need to construct a more realistic, higher-frequency (monthly) proxy for NGDP.
  - P = Official monthly CPI.
  - Y = A **"Real Activity Index"** ("Li Keqiang-style") built from indicators like electricity production, rail freight, and industrial production.
  - $NGDP_Proxy = P \times Y$. This proxy will be the variable targeted in the simulation.
- **Robustness Checks:**
  - Use other, well-measured monthly variables (like `Industrial Production`) as a simpler proxy.